Rockchip USB Gadget UAC Developer Guide

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Preface

Overview

This document provides how to use the USB Gadget UAC (USB Audio Class) Driver based on the Linux Kernel on Rockchip platforms. Rockchip platforms support both UAC1 (compatible with USB Audio Class specification 1.0) and UAC2 (compatible with USB Audio Class specification 2.0) Drivers, both of which provide basic recording and playback functionality. Additionally, Rockchip platforms offer UAC1 Legacy (requiring actual sound card support and providing playback only) and Audio Source (providing recording only but supporting up to 15 different sample rates). Developers can select the appropriate UAC Driver based on specific product requirements.

To support volume adjustment/mute functionality via physical buttons on the device side, HID control must be added. The currently released SDK does not yet support this feature. Developers can refer to the following documents to implement HID functionality:

Kernel/Documentation/usb/gadget-testing.txt (refer to section 6. HID function)

Kernel/Documentation/ABI/testing/configfs-usb-gadget-hid

Universal Serial Bus Audio Device Class Specification for Basic Audio Devices (refer to section 8 HID Support in Basic Audio Devices)

Product Versions

Chip Models	Kernel Version
All Rockchip chips (excluding MCU)	Linux-4.4 and later versions

Intended Audience

This document (this guide) is mainly intended for:

- Software development Engineers
- Technical Support Engineers

Revision History

Date	Version	Author	Modification Description
2019-03-13	V1.0	Wu Liangfeng	Initial version
2019-11-11	V1.1	Wu Liangfeng	Updated document title; added support for Linux-4.19
2020-02-19	V1.1.1	Wu Liangfeng	Added disclaimer, trademark statement, and copyright notice
2021-04-07	V1.2.0	Ren Jianing Wu Liangfeng	Added support for UAC1/UAC2 multi-sample rates, multi-channel audio, volume control, and clock synchronization
2022-05-10	V1.3.0	Wang Mingcheng	Added development guidelines for Linux-5.10; optimized descriptions for Linux-4.4/Linux-4.19
2025-04-25	V1.3.1	Wu Liangfeng	Updated supported chip names and kernel versions
2026-06-22	V1.4.0	Wu Liangfeng	Corrected typos and technical errors

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1. Kernel UAC CONFIG

1.1 Related Kernel Commits

The UAC1/UAC2 kernel code in Kernel-5.10 must be updated to the following commit to support full functionality.

```
usb: gadget: u_audio: add uevent for ppm compensation
```

To properly utilize the UAC1/UAC2 features in Kernel 4.4/4.19, first verify that the kernel in use includes the following series of patches. If the patches are missing, please submit an issue to the Rockchip Redmine platform or contact the author of this document via email.

- Patch list:

```
usb: gadget: f_uac1: set baInterfaceNr of ac_header_desc dynamically
usb: gadget: u_audio: fix dev_dbg information
usb: gadget: f_uac: fix volume control to uevent
usb: gadget: u_audio: disable eps when usb disconnect
usb: gadget: u_audio: add uevent for ppm compensation
usb: gadget: f_uac: update maxpacket in function bind
usb: gadget: f_uac: add pktsize calculating at setting playback srate
usb: gadget: f_uac: add volume and mute feature unit
UPSTREAM: usb: gadget: f_uac2: fix packet size calculation
usb: gadget: u_audio: fix the risk of strcpy out of bounds
usb: gadget: f_uac1: disable IN/OUT ep if unused
usb: gadget: u_audio: add uevent for set_alt and set_srate
FROMLIST: usb: gadget: f_uac*: Support multiple sampling rates
FROMLIST: usb: gadget: f_uac*: Reduce code duplication
FROMLIST: usb: gadget: f_uac1: Fix endpoint reading
```

1.2 Related CONFIGs

CONFIG_USB_CONFIGFS_F_UAC1 (enable UAC1 Function)

CONFIG_USB_CONFIGFS_F_UAC2 (enable UAC2 Function)

CONFIG_USB_CONFIGFS_F_UAC1_LEGACY (enable UAC1 Legacy Function)

CONFIG_USB_CONFIGFS_F_ACC (Audio Source depends on it)

CONFIG_USB_CONFIGFS_F_AUDIO_SRC (enable Audio Source Function)

1.3 Related Documents

- Documentation/usb/gadget_configfs.txt
- Documentation/usb/gadget-testing.txt
- Documentation/ABI/testing/configfs-usb-gadget-uac1
- Documentation/ABI/testing/configfs-usb-gadget-uac1_legacy
- Documentation/ABI/testing/configfs-usb-gadget-uac2

1.4 Support List

Table 1-1 Features Supported by UAC1/UAC2 Device on Rockchip Platforms

Function	Uevent Control	Multi-Sample Rate	Multi-Channel	Feature Unit Volume Control	Clock Synchronization
UAC1	YES	YES	YES	YES	YES
UAC2	YES	YES	YES	YES	YES

Note:

[1] Uevent control refers to the kernel sending events such as UAC enable/disable, sample rate configuration, and volume adjustment to the userspace uac_app via uevent, thereby achieving sound card control. For details, see section Uevent.

[2] The Feature Unit volume control of UAC2 is incompatible with the Windows platform. Therefore, it is recommended to enable Feature Unit volume control for UAC1 and disable it for UAC2. For usage instructions, refer to section UAC Usage.

[3] Clock synchronization refers to dynamically adjusting the audio clock frequency to synchronize the USB host clock with the USB audio device clock. Currently, only RV1126-Linux-4.19 in the SDK supports clock synchronization. For usage instructions, refer to section PPM Compensation.

2. UAC1/UAC2 Usage and Test

2.1 UAC1 Information

USB Audio Class 1 standard (1998)

- This standard allows for a maximum of 24 bits/96 kHz.
- The standard itself does not impose any limitation on sample rate.

Class 1 is tied to USB 1 Full Speed = 12 Mbps

- A packet is sent every millisecond.

The maximum packet size is 1023 bytes. 2 channels × 24 bits × 96000 Hz sample rate = 4,608,000 bits/s or 576 bytes/ms. This fits within the 1023-byte limit. Any higher common sample rate, e.g., 176 kHz, requires 1056 bytes, exceeding the maximum packet size.

- All operating systems (Windows, macOS, and Linux) support USB Audio Class 1 natively.

This means no driver installation is required; it is plug-and-play. All support 2-channel audio with 24-bit words and a 96 kHz sample rate.

The Rockchip platform UAC1 Driver supports the USB Audio Class Specification 1.0, enabling both recording and playback, and **does not require a physical sound card**.

The UAC1 Driver sets bInterval=4.

Default support:

- Speed: High Speed
- Sample Rate: 48 kHz for both playback and capture
- Channel Count: 2 Channels for both playback and capture
- Bit Depth: 16 bits for both playback and capture

Note:

*[1] The USB Audio 1.0 Specification was completed prior to the USB 2.0 Core Specification; therefore, the USB Audio 1.0 Specification does not include the concept of High Speed mode. High Speed mode for Audio 1.0 compatible devices can be achieved on specific operating systems using certain empirical rules. For example, modifying the **bInterval=4** field in the isochronous endpoint descriptor currently enables High Speed mode support for UAC1 across various platforms. This approach offers better compatibility than UAC2 on different Windows system versions.*

[2] Sample rate, channel count, and bit depth can all be configured via interfaces provided by the kernel. For detailed usage instructions, please refer to UAC Usage.

2.2 UAC2 Information

USB Audio Class 2 Standard (2009)

- USB Audio Class 2 additionally supports 32-bit resolution and all common sample rates exceeding 96 kHz.

Class 2 operates in High-Speed mode (480 Mbps), which requires USB 2.0 or USB 3.0. Given that the High-Speed data rate is 40 times that of Full-Speed, recording 60 channels at 24-bit/96 kHz (138 Mbps) presents no issue.

- Using High Speed USB for playback, there are no limits in resolution.
- It is downwards compatible with Class 1.
- From mid-2010 onwards, USB Audio Class 2 Drivers are available in OSX 10.6.4 and Linux.

Both support sample rates up to 384 kHz.

- Microsoft simply did not support UAC2.

In April 2017, an update to Win10 finally introduced native mode Drivers.

If you use older versions of Windows, you still need a third-party Driver.

The Rockchip platform UAC2 Driver supports the USB Audio Class Specification 2.0, enabling both recording and playback, and **does not require a physical sound card**.

Default support:

- Speed: High Speed
- Sample Rate: 48 kHz for both playback and capture
- Channels: 2 Channels for both playback and capture
- Bit Depth: 16 bits for both playback and capture

Note:

[1] Windows has supported UAC 2.0 Drivers by default only since Windows 10 (version 1703).

[2] Windows and Linux have slightly different response workflows for audio events; compatibility handling is required. Linux behaves the same as Android in this regard.

[3] Windows caches device Driver information. After changing configurations, it is recommended to uninstall the Driver to allow Windows to re-identify the device.

[4] Sample rate, channel count, and bit depth can all be configured via kernel-provided interfaces. For detailed usage instructions, refer to UAC Usage.

2.3 UAC Usage

Currently, on Rockchip platforms, UAC1 and UAC2 are essentially identical in configuration methods and implemented functionality, differing only in compatibility and protocol specifications.

- Add defconfig

For UAC1, add `CONFIG_USB_CONFIGFS_F_UAC1=y` to the kernel defconfig.

For UAC2, add `CONFIG_USB_CONFIGFS_F_UAC2=y` to the kernel defconfig.

- Linux configfs configuration

Refer to the following scripts for configuring UAC1/UAC2 on Kernel 4.4/4.19:

```
#!/bin/sh

UAC=uac2

USB_ATTRIBUTE=0x409
USB_GROUP=rockchip
USB_SKELETON=b.1

CONFIGFS_DIR=/sys/kernel/config
USB_CONFIGFS_DIR=${CONFIGFS_DIR}/usb_gadget/${USB_GROUP}
USB_STRINGS_DIR=${USB_CONFIGFS_DIR}/strings/${USB_ATTRIBUTE}
USB_FUNCTIONS_DIR=${USB_CONFIGFS_DIR}/functions
USB_CONFIGS_DIR=${USB_CONFIGFS_DIR}/configs/${USB_SKELETON}

function_init()
{
    mkdir ${USB_FUNCTIONS_DIR}/${UAC}.gs0
}
```

```
configfs_init()
{
    echo "Debug: configfs_init"
    mkdir /dev/usb-ffs

    mount -t configfs none ${CONFIGFS_DIR}
    mkdir ${USB_CONFIGFS_DIR} -m 0770
    echo 0x2207 > ${USB_CONFIGFS_DIR}/idVendor
    echo 0x0310 > ${USB_CONFIGFS_DIR}/bcdDevice
    echo 0x0200 > ${USB_CONFIGFS_DIR}/bcdUSB
    mkdir ${USB_STRINGS_DIR}    -m 0770
    SERIAL=`cat /proc/cpuinfo | grep Serial | awk '{print $3}'`
    if [ -z $SERIAL ];then
        SERIAL=0123456789ABCDEF
    fi
    echo $SERIAL > ${USB_STRINGS_DIR}/serialnumber
    echo "rockchip"  > ${USB_STRINGS_DIR}/manufacturer
    echo "uac-test"  > ${USB_STRINGS_DIR}/product

    function_init

    mkdir ${USB_CONFIGS_DIR}  -m 0770
    mkdir ${USB_CONFIGS_DIR}/strings/${USB_ATTRIBUTE}  -m 0770
}

parameter_init()
{
    if [ "$UAC" == "uac2" ]; then
        CONFIG_STRING=uac2
        echo "parameter_init ${CONFIG_STRING}"
    elif [ "$UAC" == "uac1" ];then
        CONFIG_STRING=uac1
        echo "parameter_init ${CONFIG_STRING}"
    else
        echo "parameter_init ${UAC} is invalid"
    fi
}

config_init()
{
    UAC_GS0=${USB_FUNCTIONS_DIR}/${UAC}.gs0
    echo 3 > ${UAC_GS0}/p_chmask
    echo 2 > ${UAC_GS0}/p_ssize
    echo 8000,16000,44100,48000 > ${UAC_GS0}/p_srate
    echo 0 > ${UAC_GS0}/p_feature_unit
    echo 3 > ${UAC_GS0}/c_chmask
    echo 2 > ${UAC_GS0}/c_ssize
    echo 8000,16000,44100,48000 > ${UAC_GS0}/c_srate
    echo 0 > ${UAC_GS0}/c_feature_unit
}

syslink_function()
{
```

```
    ln -s ${USB_FUNCTIONS_DIR}/$1 ${USB_CONFIGS_DIR}/f${USB_FUNCTIONS_CNT}
    let USB_FUNCTIONS_CNT=USB_FUNCTIONS_CNT+1
}

bind_functions()
{
    USB_FUNCTIONS_CNT=1

    if [ "$UAC" == "uac2" ]; then
        syslink_function uac2.gs0
    elif [ "$UAC" == "uac1" ];then
        syslink_function uac1.gs0
    else
        echo "parameter_init ${UAC} is invalid"
    fi

    echo ${CONFIG_STRING} > ${USB_CONFIGS_DIR}/strings/${USB_ATTRIBUTE}/configuration
}

program_kill()
{
    P_PID=`ps | grep $1 | grep -v grep | awk '{print $1}'`
    test -z ${P_PID} || kill -9 ${P_PID}
}

usb_device_stop()
{
    echo "none" > ${USB_CONFIGFS_DIR}/UDC
    program_kill adbd
    program_kill mtp-server
    ls ${USB_CONFIGS_DIR} | grep f[0-9] | xargs -I {} rm ${USB_CONFIGS_DIR}/{}
}

case "$1" in
start)
    echo "usb_config in $1"
    DIR=$(cd `dirname $0`; pwd)
    parameter_init
    if [ -z $CONFIG_STRING ]; then
        echo "$0: no function be selected"
        exit 0
    fi
    test -d ${USB_CONFIGFS_DIR} || configfs_init
    echo 0x0019 > ${USB_CONFIGFS_DIR}/idProduct
    config_init
    bind_functions
    sleep 1
    UDC=`ls /sys/class/udc/| awk '{print $1}'`
    echo $UDC > ${USB_CONFIGFS_DIR}/UDC
    ;;
stop)
    usb_device_stop
    ;;
```

```
restart|reload)
    # Do restart usb by udev
    echo "USB_FORCE_CHANGED" >> /tmp/.usb_config
    usb_device_stop
    sleep 1
    $0 start
    # Don't forget to clear "USB_FORCE_CHANGED"
    sed -i "/USB_FORCE_CHANGED/d" /tmp/.usb_config
    ;;
*)
    echo "Usage: $0 {start|stop|restart}"
    exit 1
esac

exit 0
```

Note:

[1] Directly run `uac.sh restart` to switch to UAC1/UAC2

[2] Modify the variable `${UAC}` to specify UAC1 or UAC2

[3] "idProduct" can be defined according to the product, but must not conflict with other USB Function idProduct values of the product

[4] Windows caches device Drivers; after changing the configuration, it is recommended to uninstall the driver to allow Windows to re-identify the device

The Kernel-5.10 UAC1/UAC2 configuration is integrated into the `/oem/usr/bin/usb_config.sh` script:

```
[...]

uac1_device_config()
{
  UAC=$1
  mkdir ${USB_FUNCTIONS_DIR}/${UAC}.gs0
  UAC_GS0=${USB_FUNCTIONS_DIR}/${UAC}.gs0
  echo 3 > ${UAC_GS0}/p_chmask
  echo 2 > ${UAC_GS0}/p_ssize
  echo 1 > ${UAC_GS0}/p_mute_present
  echo 1 > ${UAC_GS0}/p_volume_present
  echo -5120 > ${UAC_GS0}/p_volume_min #-20db min must > -96db
  echo 8000,16000,44100,48000 > ${UAC_GS0}/p_srate

  echo 3 > ${UAC_GS0}/c_chmask
  echo 2 > ${UAC_GS0}/c_ssize
  echo 4 > ${UAC_GS0}/req_number
  echo 1 > ${UAC_GS0}/c_mute_present
  echo 1 > ${UAC_GS0}/c_volume_present
  echo -3200 > ${UAC_GS0}/c_volume_min #-12.5db
  echo 0 > ${UAC_GS0}/c_volume_max    #0db
  echo 32 > ${UAC_GS0}/c_volume_res #0.125db
  echo 8000,16000,44100,48000 > ${UAC_GS0}/c_srate
```

```
  ln -s ${UAC_GS0} ${USB_CONFIGS_DIR}/f2
}
uac2_device_config()
{
  UAC=$1
  mkdir ${USB_FUNCTIONS_DIR}/${UAC}.gs0
  UAC_GS0=${USB_FUNCTIONS_DIR}/${UAC}.gs0
  echo 3 > ${UAC_GS0}/p_chmask
  echo 2 > ${UAC_GS0}/p_ssize
  echo 1 > ${UAC_GS0}/p_mute_present
  echo 1 > ${UAC_GS0}/p_volume_present
  echo -5120 > ${UAC_GS0}/p_volume_min #-20db min must > -96db
  echo 8000,16000,44100,48000 > ${UAC_GS0}/p_srate

  echo 3 > ${UAC_GS0}/c_chmask
  echo 2 > ${UAC_GS0}/c_ssize
  echo 4 > ${UAC_GS0}/req_number
  echo 1 > ${UAC_GS0}/c_mute_present
  echo 1 > ${UAC_GS0}/c_volume_present
  echo -3200 > ${UAC_GS0}/c_volume_min #-12.5db
  echo 0 > ${UAC_GS0}/c_volume_max   #0db
  echo 32 > ${UAC_GS0}/c_volume_res #0.125db
  echo 8000,16000,44100,48000 > ${UAC_GS0}/c_srate

  ln -s ${UAC_GS0} ${USB_CONFIGS_DIR}/f2
}

[...]
```

2.3.1 Parameters

By examining the function directory for UAC1/UAC2 in Kernel-4.4/Kernel-4.19, the following parameter interfaces can be identified:

```
[root@RV1126_RV1109:/sys/kernel/config/usb_gadget/rockchip/functions/uac1.gs0]# ls
c_chmask        c_srate  p_chmask        p_srate  req_number
c_feature_unit  c_ssize  p_feature_unit  p_ssize
```

Here, 'c' denotes capture, referring to the recording channel of the USB virtual sound card, which corresponds to playback on the Host. Similarly, 'p' denotes playback, referring to the playback channel of the USB virtual sound card, which corresponds to recording on the Host. The two channels operate independently.

Common parameter interface settings are as follows:

- Sample Rate

p_srate, c_srate

The default value is a single sample rate of 48000. Multiple sample rates are supported and must be separated by commas. For example:

`echo 8000,16000,44100,48000 > ${UAC_GS0}/c_srate`

- Channel

c_chmask, p_chmask

The default value is 3 (stereo). Multi-channel configurations or disabling specific channels are supported.

For example, to configure the UAC Device for 5.1 recording channels and disable playback functionality:

`echo 0x3F > p_chmask`

`echo 0 > c_chmask`

- Data Depth

p_ssize, c_ssize. The default value is 2 (Unit: Bytes, i.e., 16-bit depth).

- Volume

c_feature_unit and p_feature_unit are switches for volume control. The default is 0 (disabled); write 1 to enable.

Since the UAC2 Feature Unit volume control function is incompatible with the Windows platform, it is recommended to enable Feature Unit volume control for UAC1 and disable it for UAC2.

- req_number

req_number specifies the number of Buffers submitted by the UAC Driver to the USB Controller.

According to the USB protocol, when the USB Device is not ready to synchronize transmission data, it is permitted to reply with a NO DATA packet; however, macOS does not support this behavior. Therefore, under high CPU load scenarios, increasing the req_number value expands the UAC Buffer cache, thereby enhancing compatibility across different operating systems.

The function directory for UAC1/UAC2 in Kernel 5.10 differs slightly from that in Kernel 4.4/Kernel 4.19. The specific parameters are as follows:

```
#ls /sys/kernel/config/usb_gadget/rockchip/functions/uac1.gs0/
c_chmask          p_ssize           p_volume_max      c_volume_max
c_srate           req_number        p_volume_res      c_volume_res
c_ssize           p_mute_present    c_mute_present    function_name
p_chmask          p_volume_present  c_volume_present
p_srate           p_volume_min      c_volume_min
```

Among them:

- The enable nodes for volume/mute have changed from `c_feature_unit` and `p_feature_unit` to `p_mute_present` / `p_volume_present` and `c_mute_present` / `c_volume_present`. The kernel enables volume/mute by default, so the SDK solution `uac_app` does not need to control them by default.
- The MIN_DB/MAX_DB/RES_DB values for volume/mute have been changed to -100 dB/0 dB/1 dB, respectively. Meanwhile, the kernel provides the nodes `p_volume_max` / `p_volume_min` / `p_volume_res` and `c_volume_max` / `c_volume_min` / `c_volume_res`, allowing product developers to reconfigure these parameters according to codec characteristics.

Note:

[1] volume/mute requires adaptation in `uac_app` to control codec output more linearly based on the PC volume slider. For reference, see Windows Audio-Tapered Volume Controls.

[2] The MIN_DB/MAX_DB values defined on the device side will be reported to the PC during UAC enumeration. The PC will then map 0 and 100 on its volume slider to MIN_DB and MAX_DB, respectively.

2.3.2 Uevent

To accurately control audio data flow, as well as configure multiple sample rates and volume levels, the user-space application `uac_app` must receive and respond to kernel events. Here, we use uevent events to notify the user space from the kernel. The application monitors the UAC node and extracts key string information from the events.

start/stop

```
g_audio_work: sent uac uevent USB_STATE=SET_INTERFACE STREAM_DIRECTION=IN STREAM_STATE=OFF
g_audio_work: sent uac uevent USB_STATE=SET_INTERFACE STREAM_DIRECTION=OUT STREAM_STATE=ON
```

multi-sample

```
g_audio_work: sent uac uevent USB_STATE=SET_SAMPLE_RATE STREAM_DIRECTION=OUT
SAMPLE_RATE=44100
```

volume/mute

```
g_audio_work: sent uac uevent USB_STATE=SET_VOLUME STREAM_DIRECTION=IN VOLUME=-1154
g_audio_work: sent uac uevent USB_STATE=SET_VOLUME STREAM_DIRECTION=IN VOLUME=-350
g_audio_work: sent uac uevent USB_STATE=SET_VOLUME STREAM_DIRECTION=IN VOLUME=0
g_audio_work: sent uac uevent USB_STATE=SET_VOLUME STREAM_DIRECTION=IN VOLUME=1739
g_audio_work: sent uac uevent USB_STATE=SET_VOLUME STREAM_DIRECTION=IN VOLUME=2612
```

Note:

[1] Here, IN indicates Host recording, and OUT indicates Host playback.

[2] For Kernel-4.4/Kernel-4.19, the UAC1/UAC2 Driver automatically converts the raw Volume control value set by the PC into a Volume percentage and reports it to uac_app via Uevent.

[3] In Kernel-5.10, the raw Volume control value set by the PC is reported directly. The uac_app must convert the value reported by the kernel into the corresponding dB value for the specific codec. The conversion formula is as follows:

```
// Conversion formula between volume control and dB
#define VOLUME_USB_TO_DB(v_usb) (v_usb <= 0x7FFF) ? v_usb : (-(((int16_t)0xFFFF -
(int16_t)v_usb) + 1))
```

Method used by uac_app to convert dB to volume P: Percentage; a: current volume in decibels; m: maximum volume in decibels P = 10 ^ [(a - m)/10] a = 10 * lg(P) + m

2.3.3 PPM Compensation

Since the USB clock originates from the Host and the Codec clock originates from the Device, clock asynchrony can lead to audio data loss, manifested as periodic underrun/overrun messages printed by ALSA.

To address the issue of asynchronous clocks, we propose a PPM compensation solution. This method calculates the PPM deviation over a specific period by comparing the system time with the SOF packets sent by the Host every 125 μs, and then reports the result to the application layer via a uevent:

```
g_audio_work: sent uac uevent USB_STATE=SET_AUDIO_CLK PPM=12
g_audio_work: sent uac uevent USB_STATE=SET_AUDIO_CLK PPM=-1
```

As long as the USB connection is established, the ppm value is continuously calculated. The application layer listens to uevent events to obtain the specific ppm value and then adjusts the codec clock via the interfaces provided by the codec.

Note:

The ppm value measured using this method becomes increasingly stable over time; generally, no new ppm values are reported after ten minutes. If the USB connection is interrupted, SOF counting restarts, causing the ppm value to reset to 0 before gradually stabilizing at a new value.

If real-time performance is not critical, the application layer may ignore the reported ppm compensation value.

2.3.4 Compatibility

Volume Control

When using UAC2 on Windows systems, volume control must be disabled. Specifically, set c_feature_unit and p_feature_unit of UAC2 to 0; otherwise, Windows fails to load the UAC2 driver properly.

Table 2-1 Support for UAC Volume Control Across Different Host Systems

	Windows 10	Linux (Ubuntu)	MacOS
UAC1	Supported	Unsupported	Supported
UAC2	Driver Error	Unsupported	Supported

2.4 UAC Test

The testing methods for UAC1 and UAC2 are identical; UAC1 is used here as an example.

After successfully executing the configuration script, connect the USB device to a PC. The PC should recognize the USB Audio device, as shown in Figure 2-1 (Windows-USB-Audio-Class1), Figure 2-2 (Ubuntu-USB-Audio-Class1-Output), and Figure 2-3 (Ubuntu-USB-Audio-Class1-Input).



Figure 2-1 Windows USB Audio Class 1



Figure 2-2 Ubuntu-USB-Audio-Class1-Output

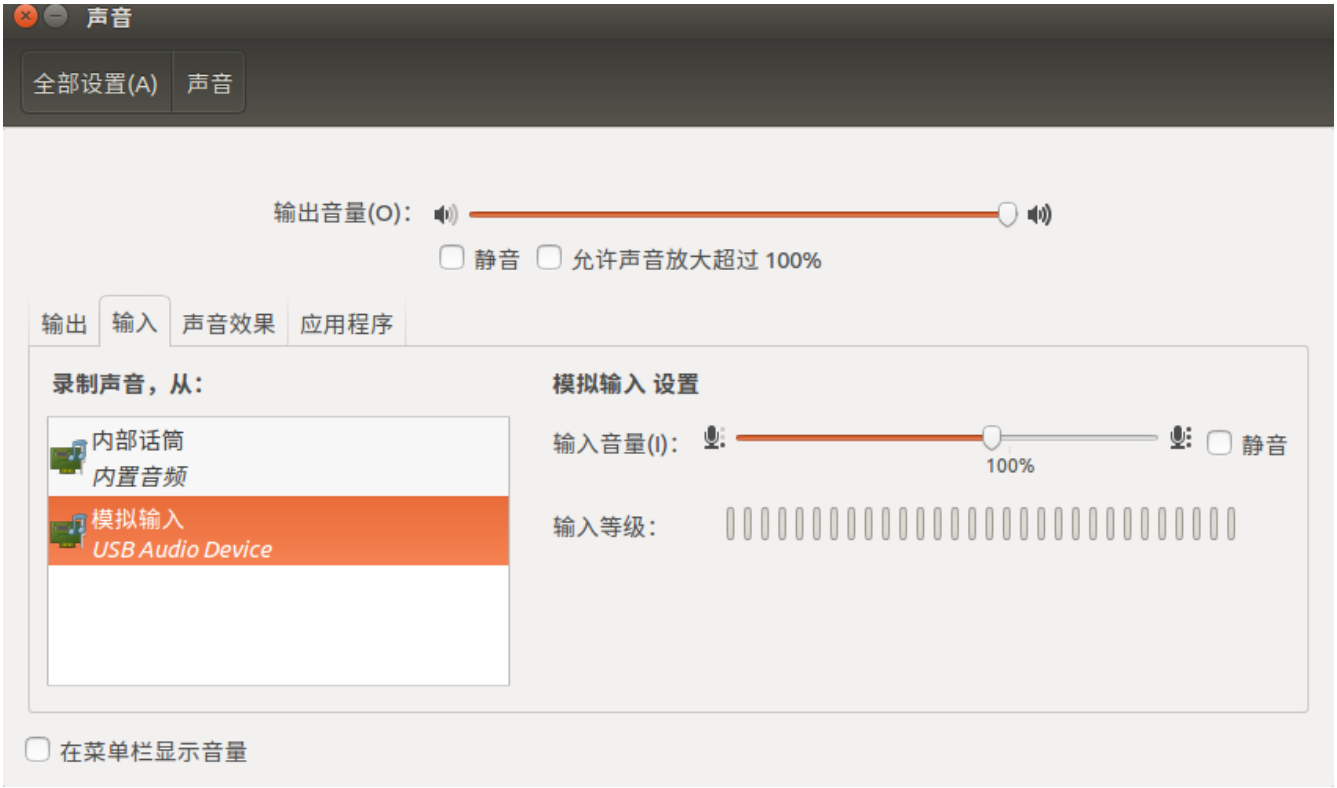


Figure 2-3 Ubuntu-USB-Audio-Class1-Input

The following serial console output from the RK3308 shows logs indicating successful enumeration of USB UAC1:

```
dwc2 ff400000.usb: new device is high-speed
dwc2 ff400000.usb: new address 19
android_work: sent uevent USB_STATE=CONNECTED
configfs-gadget gadget: high-speed config #1: b
android_work: sent uevent USB_STATE=CONFIGURED
```

To view UAC1 sound card information:

The output shown below indicates that UAC1 corresponds to card2 (UAC1Gadget), which has one playback device node, pcmC2D0p, and one capture device node, pcmC2D0c.

```
# cat /proc/asound/cards
 0 [rockchiprk3308v]: rockchip_rk3308 - rockchip,rk3308-vad
                      rockchip,rk3308-vad
 1 [rockchiprk3308p]: rockchip_rk3308 - rockchip,rk3308-pcm
                      rockchip,rk3308-pcm
 2 [UAC1Gadget     ]: UAC1_Gadget - UAC1_Gadget
                      UAC1_Gadget 0
 7 [Loopback       ]: Loopback - Loopback
                      Loopback 1

# ls -lh /proc/asound/card2
-r--r--r--    1 root     root           0 Dec 31 19:14 id
dr-xr-xr-x    3 root     root           0 Dec 31 19:14 pcm0c
dr-xr-xr-x    3 root     root           0 Dec 31 19:14 pcm0p
```

```
# ls /dev/snd/
controlC0  controlC7  pcmC1D0c   pcmC2D0p   pcmC7D1c
controlC1  pcmC0D0c   pcmC1D0p   pcmC7D0c   pcmC7D1p
controlC2  pcmC0D0p   pcmC2D0c   pcmC7D0p   timer
```

2.4.1 UAC Test on Windows

Open the Windows Sound settings, as shown in Figure 2-4 Windows-Audio-Setting, and select USB-Audio as both the audio output device and the audio input device (microphone).

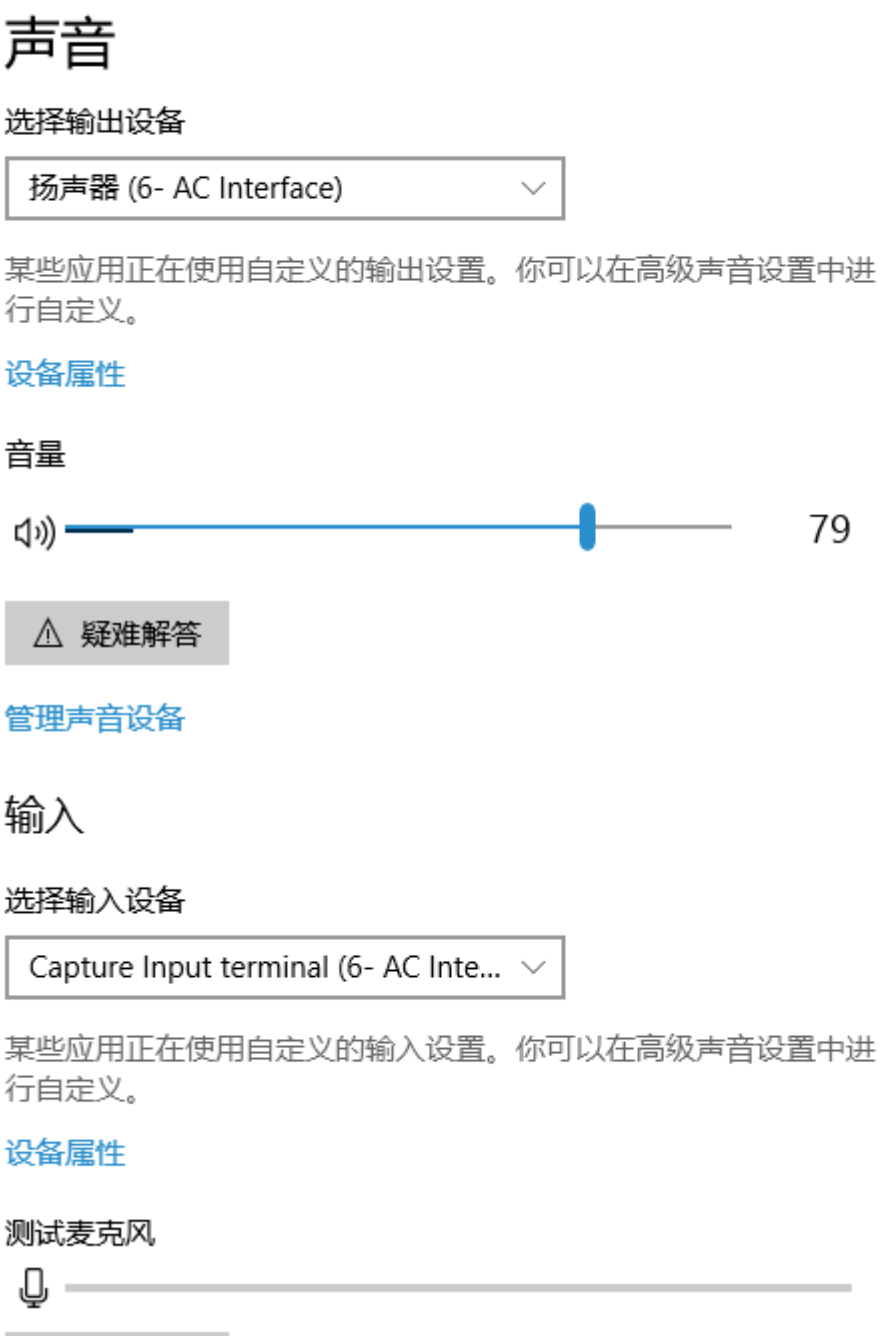


Figure 2-4 Windows Audio Settings

Test Windows UAC1 Playback Function:

Execute the following UAC1 playback command on the RK3308 serial console:

```
arecord -f dat -t wav -r 48000 -c 2 -D hw:2,0 | aplay -f dat -r 48000 -c 2 -D hw:0,0
```

The preceding command records audio from Card2 (USB Audio) and plays it back through the local sound card Card0.

After executing the command, when music is played on the Windows PC, the RK3308 local sound card will play the audio in real time.

Testing Windows UAC1 Recording Function:

To test the recording function, a playable audio file is required. Use either ADB push or arecord to save the test audio file (requirements: 48kHz, 2 channels, 16 bits).

For example, execute the arecord command via the serial console on the RK3308 to save the test audio file as test.wav.

```
arecord -f dat -t wav -r 48000 -c 2 -D hw:2,0 /tmp/test.wav
```

After successfully saving the audio file test.wav, execute the following UAC1 recording command:

```
aplay /tmp/test.wav -c 2 -r 48000 -D hw:2,0
```

After executing the above command, the PC can use the built-in Windows "Voice Recorder" application to save recording files, as shown in Figure 2-5 Windows-Voice-Recorder.



Figure 2-5 Windows-Voice-Recorder

In addition to the recording test methods described above, you can also use the Windows recording listen feature to play back recorded audio in real time. Proceed as follows:

Open "Sound Settings" --> "Sound Control Panel" --> "Recording" --> "Properties" --> "Listen", check "Listen to this device", and select the playback speaker.

Refer to Figure 2-6 Windows-Capture-Listen-1 and Figure 2-7 Windows-Capture-Listen-2 below.

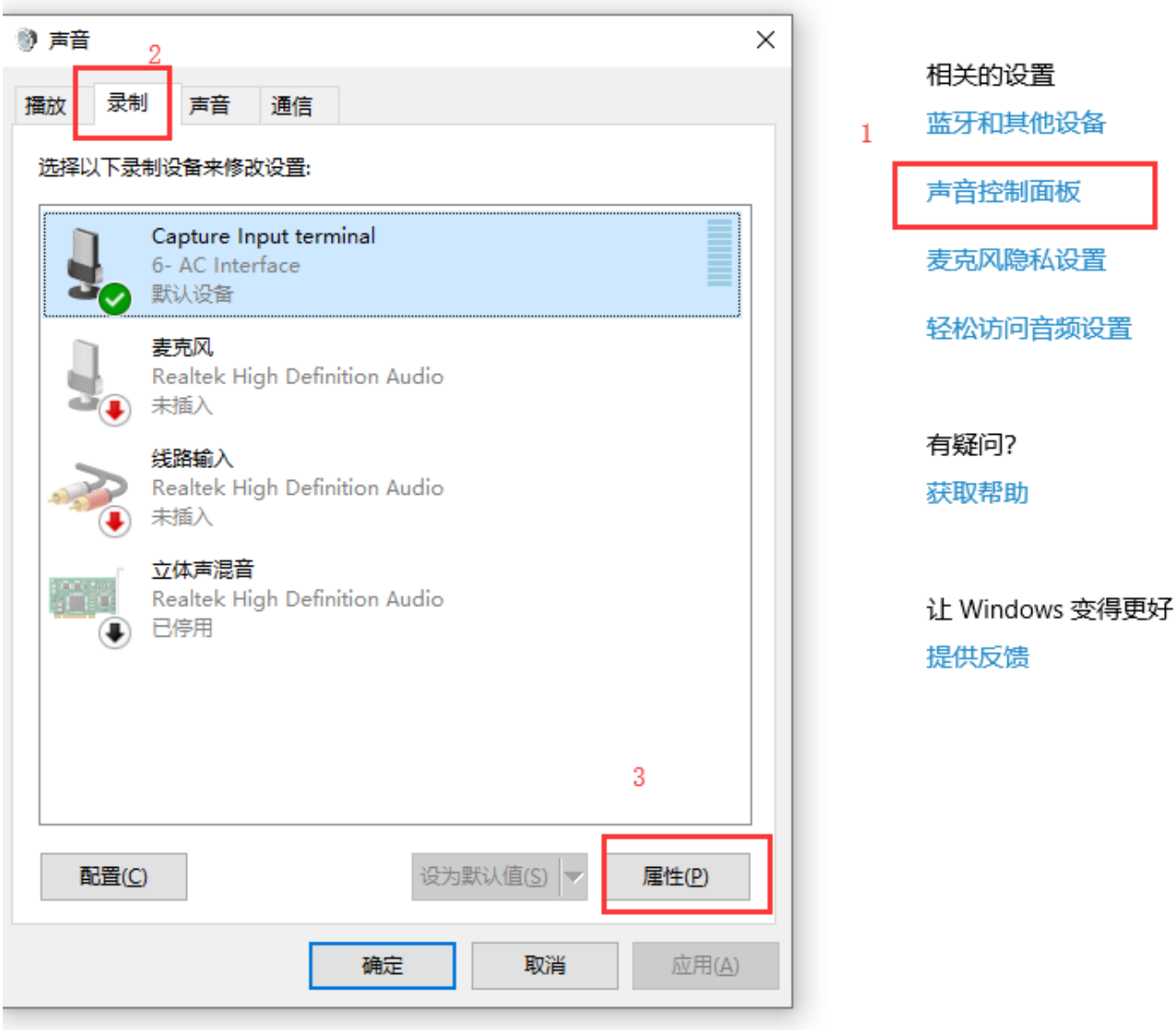


Figure 2-6 Windows-Capture-Listen-1

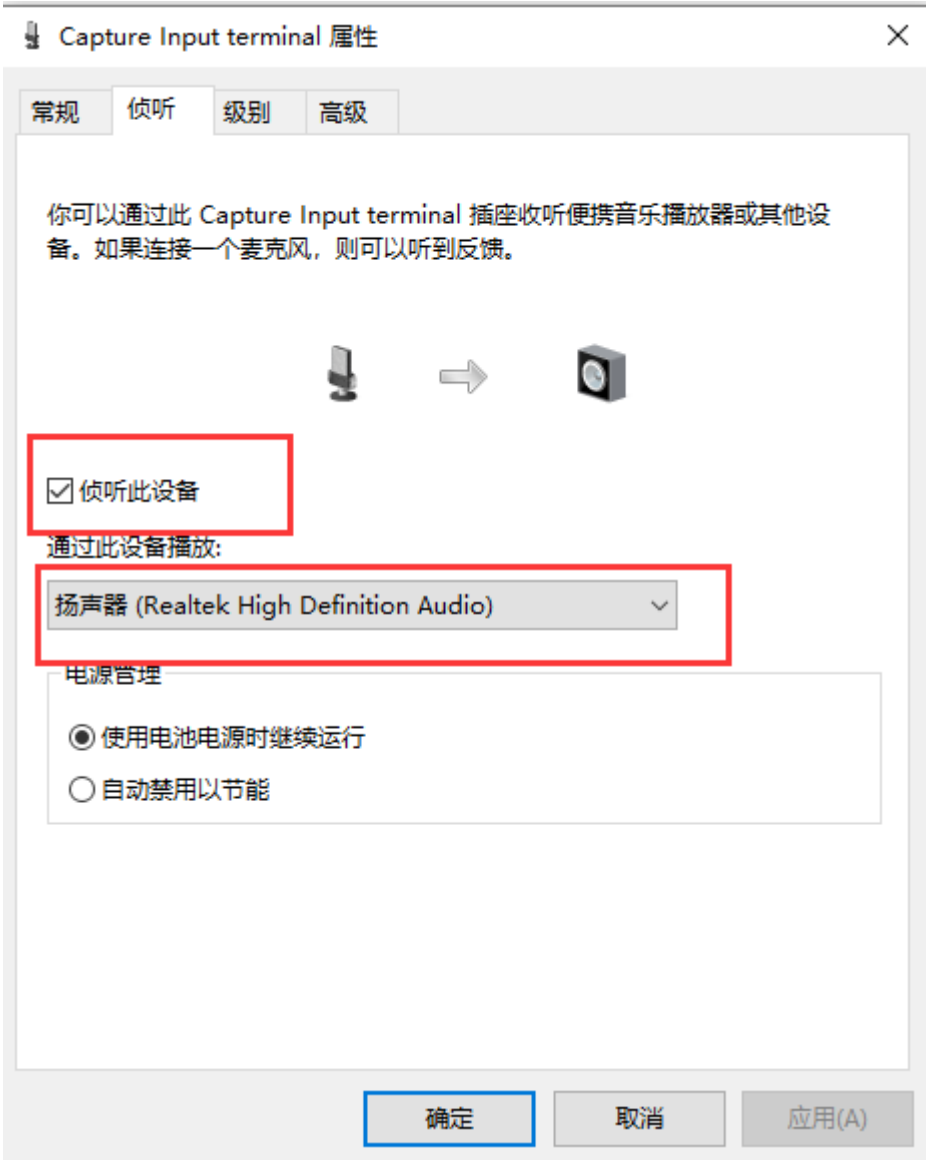


Figure 2-7 Windows-Capture-Listen-2

2.4.2 UAC Test on Ubuntu

Open the Ubuntu sound settings, as shown in Figure 2-8 Ubuntu-Audio-Setting-Output and Figure 2-9 Ubuntu-Audio-Setting-Input. Select USB-Audio as the sound output device and the sound input device (microphone), respectively.

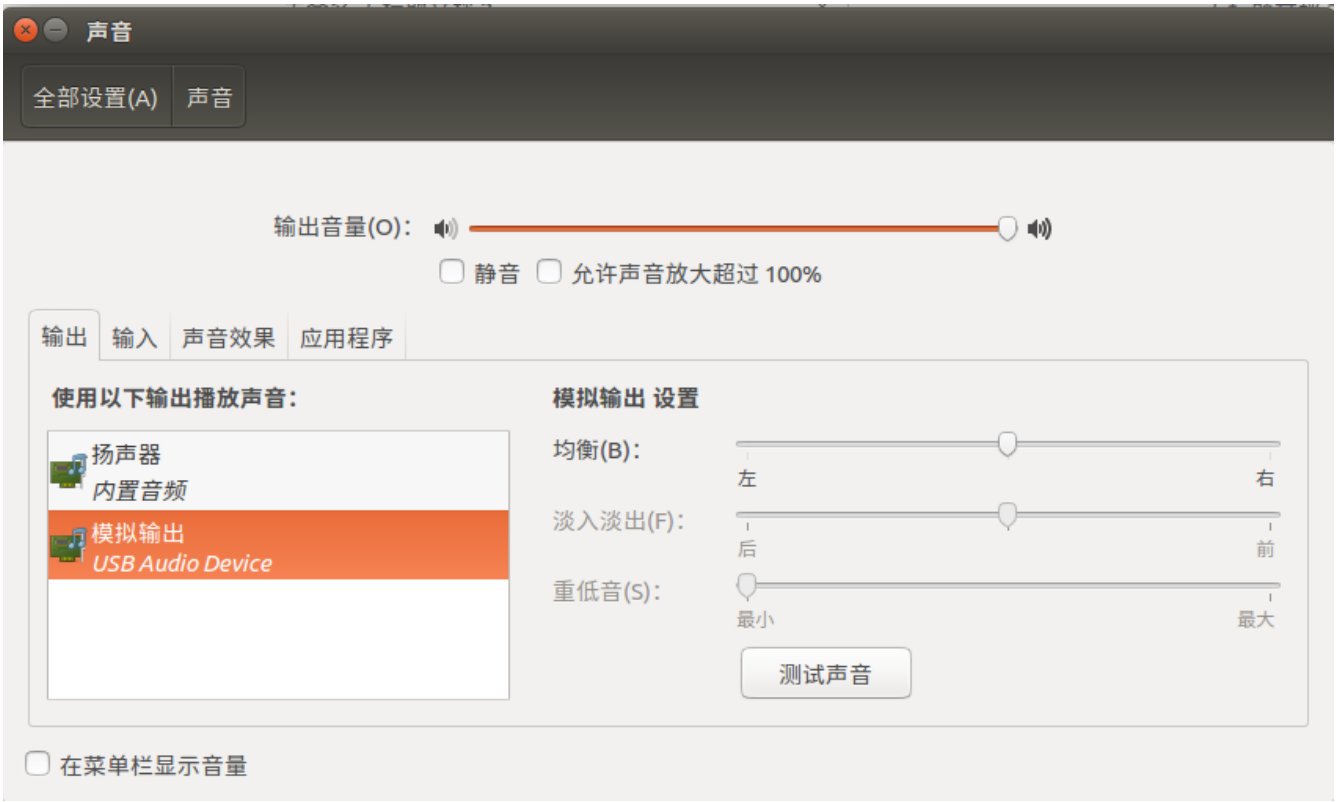


Figure 2-8 Ubuntu-Audio-Setting-Output

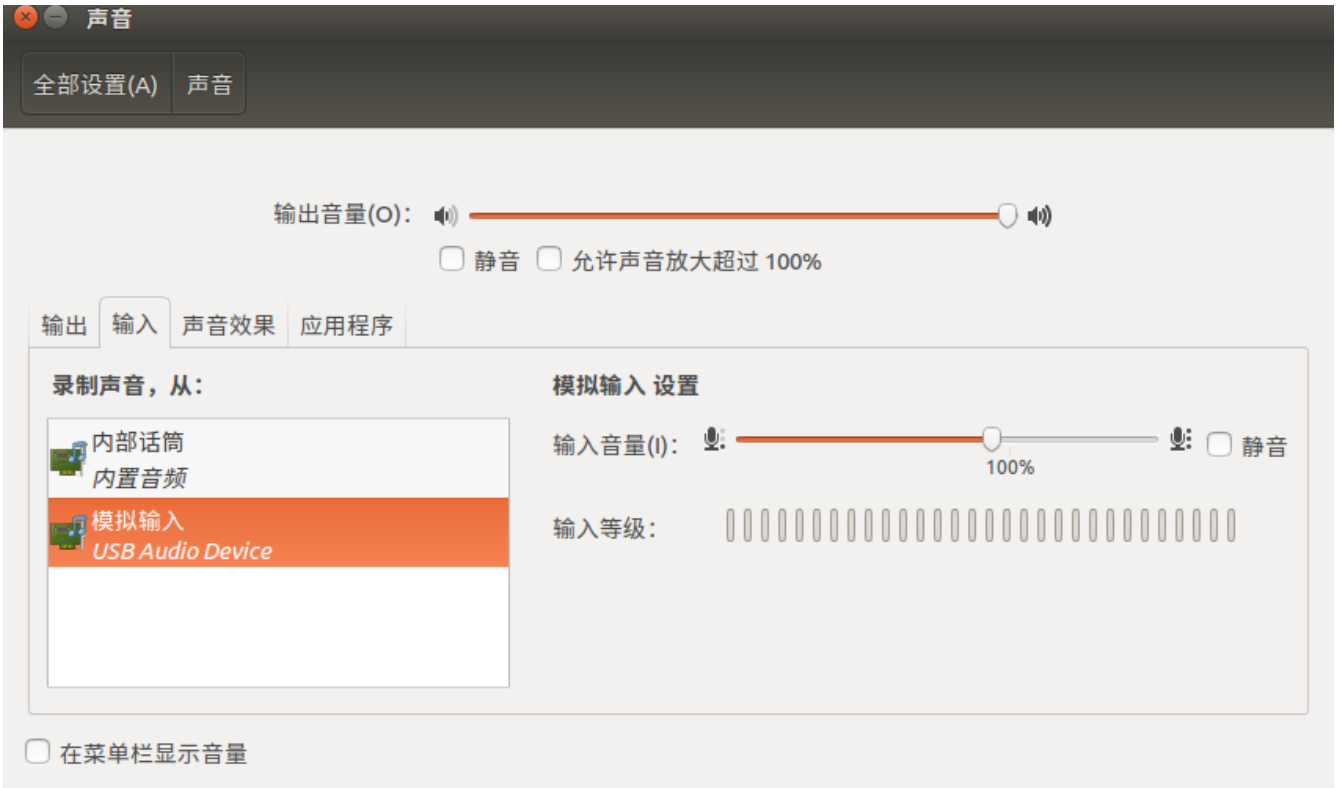


Figure 2-9 Ubuntu-Audio-Setting-Input

Test Ubuntu UAC1 Playback Function:

Execute the following UAC1 playback command on the RK3308 serial console:

```
arecord -f dat -t wav -r 48000 -c 2 -D hw:2,0 | aplay -f dat -r 48000 -c 2 -D hw:0,0
```

The preceding command records audio from Card2 (USB Audio) and plays it back through the local sound card Card0.

After executing the command, when music is played on the Ubuntu PC, the RK3308 local sound card will play the audio in real time.

Testing Ubuntu UAC1 Recording Function:

To test the recording function, a playable audio file is required. Use either ADB push or arecord to save the test audio file (requirements: 48kHz, 2 channels, 16 bits).

For example, execute the arecord command via the serial console on the RK3308 to save the test audio file as test.wav.

```
arecord -f dat -t wav -r 48000 -c 2 -D hw:2,0 /tmp/test.wav
```

After successfully saving the audio file test.wav, execute the following UAC1 recording command:

```
aplay /tmp/test.wav -c 2 -r 48000 -D hw:2,0
```

After executing the above commands, open recording software such as "Audacity" on the Ubuntu side to test the recording function.

Audacity installation command:

```
sudo apt install audacity
```

The Audacity recording interface is shown in Figure 2-10 below.

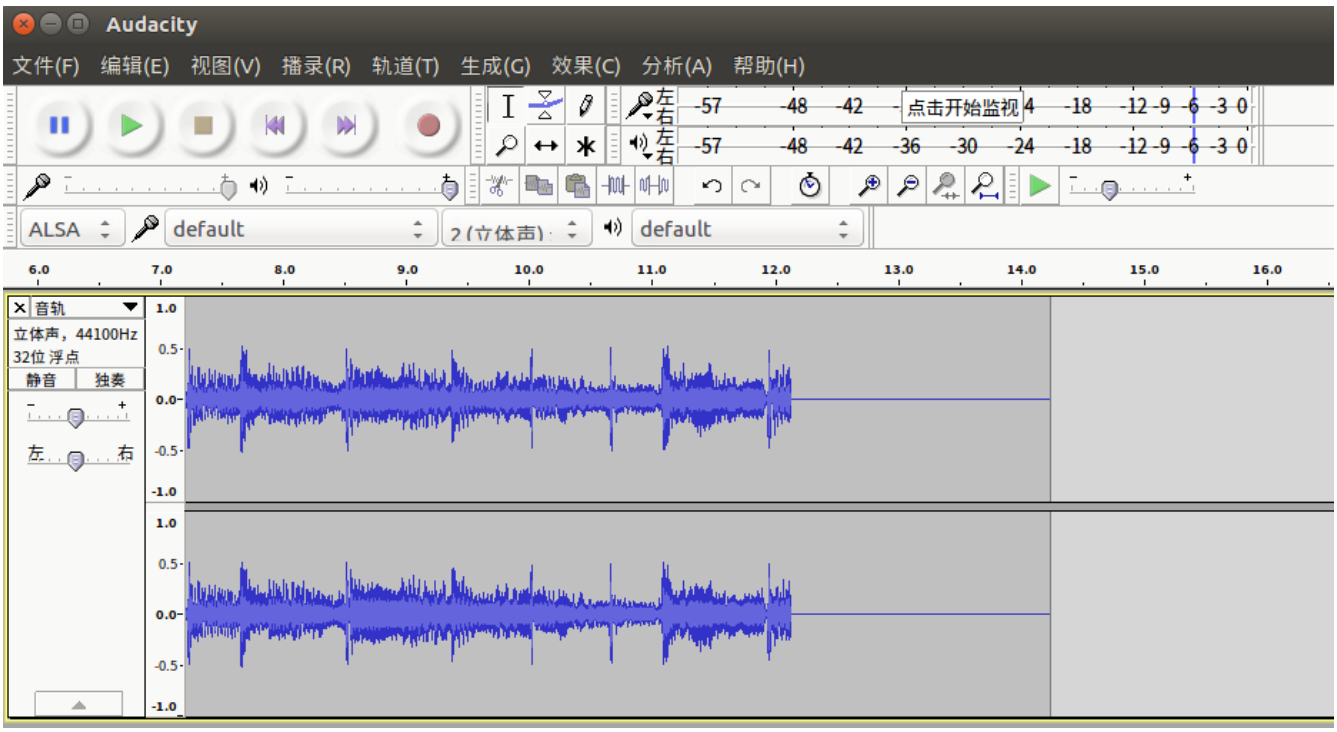


Figure 2-10 Ubuntu-audacity

3. UAC1 Legacy Usage and Test

3.1 UAC1 Legacy Usage

The Rockchip platform UAC1 Legacy Driver is compatible with the USB Audio Class Specification 1.0, but supports playback only and **requires actual sound card support (defaulting to /dev/snd/pcmC0D0p)**.

Default support:

- Speed: High Speed
- Sample Rate: playback 48 KHz, not configurable
- Channels: playback 2 Channels, not configurable
- Bit Depth: playback 16 bits

Usage of UAC1 Legacy is as follows:

Add CONFIG_USB_CONFIGFS_F_UAC1_LEGACY=y to the kernel defconfig.

Taking RK3308 EVB as an example:

Refer to the following script for configuring UAC1 Legacy:

```
mount -t configfs none /sys/kernel/config
mkdir /sys/kernel/config/usb_gadget/rockchip  -m 0770
echo 0x2207 > /sys/kernel/config/usb_gadget/rockchip/idVendor
echo 0x0019 > /sys/kernel/config/usb_gadget/rockchip/idProduct
echo 0x0100 > /sys/kernel/config/usb_gadget/rockchip/bcdDevice
mkdir /sys/kernel/config/usb_gadget/rockchip/strings/0x409   -m 0770
echo "0123456789ABCDEF" > /sys/kernel/config/usb_gadget/rockchip/strings/0x409/serialnumber
echo "rockchip"  > /sys/kernel/config/usb_gadget/rockchip/strings/0x409/manufacturer
echo "USB Audio Device"  > /sys/kernel/config/usb_gadget/rockchip/strings/0x409/product
mkdir /sys/kernel/config/usb_gadget/rockchip/configs/b.1  -m 0770
mkdir /sys/kernel/config/usb_gadget/rockchip/configs/b.1/strings/0x409  -m 0770
echo 500 > /sys/kernel/config/usb_gadget/rockchip/configs/b.1/MaxPower
echo "uac1" > /sys/kernel/config/usb_gadget/rockchip/configs/b.1/strings/0x409/configuration
mkdir /sys/kernel/config/usb_gadget/rockchip/functions/uac1_legacy.gs0
ln -s /sys/kernel/config/usb_gadget/rockchip/functions/uac1_legacy.gs0
/sys/kernel/config/usb_gadget/rockchip/configs/b.1/uac1_legacy.gs0
echo ff400000.usb > /sys/kernel/config/usb_gadget/rockchip/UDC
```

If the ADB configuration script runs by default after the RK3308 boots, the aforementioned configuration method will fail. During the debugging phase, you can manually execute the following commands to configure the UAC1 Legacy function. For the final product, the USB configuration script must integrate the ADB and UAC1 Legacy configuration scripts according to actual requirements.

```
rm -rf /sys/kernel/config/usb_gadget/rockchip/configs/b.1/ffs.adb

mkdir /sys/kernel/config/usb_gadget/rockchip/functions/uac1_legacy.gs0
echo 0x0019 > /sys/kernel/config/usb_gadget/rockchip/idProduct
echo 0x0100 > /sys/kernel/config/usb_gadget/rockchip/bcdDevice
echo "USB Audio Device" > /sys/kernel/config/usb_gadget/rockchip/strings/0x409/product
echo "uac1" > /sys/kernel/config/usb_gadget/rockchip/configs/b.1/strings/0x409/configuration
cd /sys/kernel/config/usb_gadget/rockchip/configs/b.1
ln -s ../../functions/uac1_legacy.gs0

echo ff400000.usb > ../../UDC
```

Note:

- "idProduct" can be defined according to the product requirements, but must not conflict with the idProduct of other USB Functions on the product.
- "UDC" refers to the USB controller name, corresponding to /sys/class/udc/<controller_name>.
- Windows caches device Drivers. After modifying the configuration, it is recommended to uninstall the driver to allow Windows to re-enumerate the device.

After the configuration script executes successfully, connect the USB port to a PC. The PC will recognize the USB Audio device, as shown in Figure 3-1.



Figure 3-1 Windows-USB-Audio-Class1-Legacy

The following UART log from the RK3308 side shows successful enumeration of USB UAC1 Legacy:

```
configfs-gadget gadget: Hardware params: access 3, format 2, channels 2, rate 48000
dwc2 ff400000.usb: bound driver configfs-gadget
dwc2 ff400000.usb: new device is high-speed
dwc2 ff400000.usb: new address 25
android_work: sent uevent USB_STATE=CONNECTED
configfs-gadget gadget: high-speed config #1: b
android_work: sent uevent USB_STATE=CONFIGURED
```

The UAC1 Legacy Driver provides the following configuration interfaces:

The following **configurations cannot take effect dynamically**, meaning they must be added to the UAC configuration script for execution.

```
# ls -lh /sys/kernel/config/usb_gadget/rockchip/functions/uac1_legacy.gs0/
-rw-r--r--    1 root     root        4.0K Dec 31 19:08 audio_buf_size
-rw-r--r--    1 root     root        4.0K Dec 31 19:08 fn_cap
-rw-r--r--    1 root     root        4.0K Dec 31 19:08 fn_cntl
-rw-r--r--    1 root     root        4.0K Dec 31 19:08 fn_play
-rw-r--r--    1 root     root        4.0K Dec 31 19:08 req_buf_size
-rw-r--r--    1 root     root        4.0K Dec 31 19:08 req_count

audio_buf_size - Audio buffer size; default is 48000.
fn_cap - Capture PCM device file name; default is /dev/snd/pcmC0D0c
fn_cntl - Control device file name, default set to /dev/snd/controlC0
fn_play - Playback PCM device file name; default is set to /dev/snd/pcmC0D0p
req_buf_size - ISO OUT endpoint request buffer size; default is 200.
req_count - ISO OUT endpoint request count; default is 256.
```

UAC1 Legacy does not create the corresponding sound card device node on the RK3308 side.

3.2 UAC1 Legacy Test

For playback settings on the Windows PC, refer to UAC Test on Windows.

For playback settings on the Ubuntu PC, refer to UAC Test on Ubuntu.

No commands need to be executed on the RK3308 side. After connecting the USB to the PC, the UAC1 Legacy Driver will by default enable playback on the local RK3308 Card0 sound card.

4. Audio Source Usage and Test

4.1 Audio Source Usage

The Rockchip platform Audio Source Driver is compatible with the USB Audio Class specification 1.0 but supports only recording functionality.

Default support:

- Speed: High Speed
- Number of channels: 2 channels, not configurable
- Bit depth: 16 bits
- Sample Rate: Defaults to 44.1KHz

A total of 15 different sample rates are supported and can be dynamically configured on the PC side.

```
8000, 11025, 16000, 22050, 24000,
32000, 40000, 44100, 48000, 56000,
64000, 72000, 80000, 88200, 96000,
```

Usage of Audio Source:

Add `CONFIG_USB_CONFIGFS_F_ACC=y` (Audio Source depends on it) to the kernel defconfig.

Add `CONFIG_USB_CONFIGFS_F_AUDIO_SRC=y` to the kernel defconfig.

Taking RK3308 EVB as an example:

Refer to the following script for configuring Audio Source:

```
mount -t configfs none /sys/kernel/config
mkdir /sys/kernel/config/usb_gadget/rockchip  -m 0770
echo 0x2207 > /sys/kernel/config/usb_gadget/rockchip/idVendor
echo 0x0019 > /sys/kernel/config/usb_gadget/rockchip/idProduct
mkdir /sys/kernel/config/usb_gadget/rockchip/strings/0x409   -m 0770
echo "0123456789ABCDEF" > /sys/kernel/config/usb_gadget/rockchip/strings/0x409/serialnumber
echo "rockchip"  > /sys/kernel/config/usb_gadget/rockchip/strings/0x409/manufacturer
echo "USB Audio Device"  > /sys/kernel/config/usb_gadget/rockchip/strings/0x409/product
mkdir /sys/kernel/config/usb_gadget/rockchip/configs/b.1  -m 0770
mkdir /sys/kernel/config/usb_gadget/rockchip/configs/b.1/strings/0x409  -m 0770
echo 500 > /sys/kernel/config/usb_gadget/rockchip/configs/b.1/MaxPower
echo "audio" > /sys/kernel/config/usb_gadget/rockchip/configs/b.1/strings/0x409/configuration
mkdir /sys/kernel/config/usb_gadget/rockchip/functions/audio_source.gs0
ln -s /sys/kernel/config/usb_gadget/rockchip/functions/audio_source.gs0
/sys/kernel/config/usb_gadget/rockchip/configs/b.1/audio_source.gs0
echo ff400000.usb > /sys/kernel/config/usb_gadget/rockchip/UDC
```

If the ADB configuration script runs by default after the RK3308 boots, the aforementioned configuration method will fail. During the debugging phase, you can manually execute the following commands to configure the Audio Source function. For the final product, the USB configuration script must integrate the ADB and Audio Source scripts according to actual requirements.

```
rm -rf /sys/kernel/config/usb_gadget/rockchip/configs/b.1/ffs.adb

mkdir /sys/kernel/config/usb_gadget/rockchip/functions/audio_source.gs0
echo 0x0019 > /sys/kernel/config/usb_gadget/rockchip/idProduct
echo 0x0100 > /sys/kernel/config/usb_gadget/rockchip/bcdDevice
echo "USB Audio Device" > /sys/kernel/config/usb_gadget/rockchip/strings/0x409/product
echo "audio" > /sys/kernel/config/usb_gadget/rockchip/configs/b.1/strings/0x409/configuration
cd /sys/kernel/config/usb_gadget/rockchip/configs/b.1
ln -s ../../functions/audio_source.gs0

echo ff400000.usb > ../../UDC
```

Note:

- `idProduct` can be defined according to the product requirements, but must not conflict with the `idProduct` of other USB functions on the product.

- `UDC` refers to the USB controller name, corresponding to `/sys/class/udc/<controller_name>`.
- Windows caches device drivers. After modifying the configuration, it is recommended to uninstall the driver to allow Windows to re-enumerate the device.

After the configuration script executes successfully, connect the USB port to a PC. The PC will recognize the USB Audio device, as shown in Figure 4-1 Windows-USB-Audio-Source and Figure 4-2 Ubuntu-USB-Audio-Source.



Figure 4-1 Windows-USB-Audio-Source

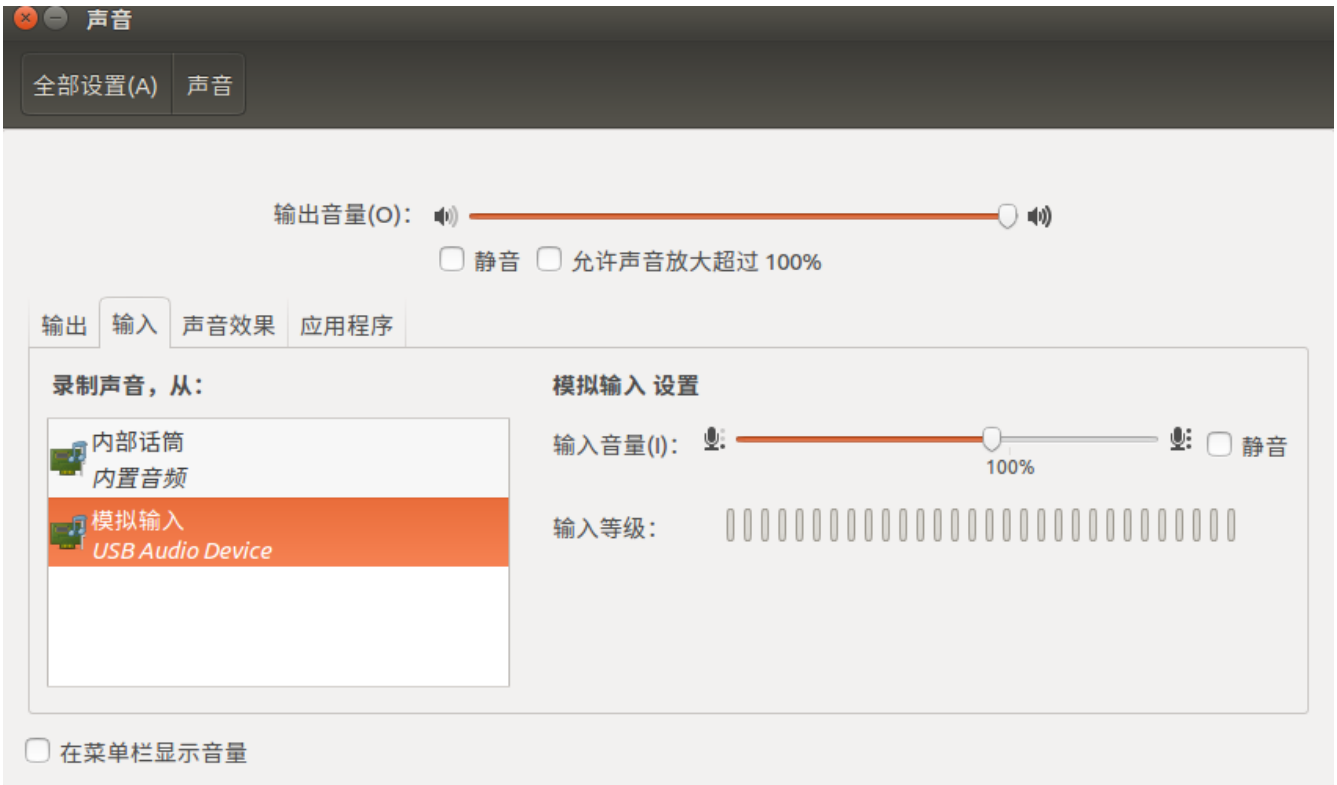


Figure 4-2 Ubuntu-USB-Audio-Input

The following serial console logs from the RK3308 side show the normal enumeration of the USB Audio Source:

```
dwc2 ff400000.usb: new device is high-speed
dwc2 ff400000.usb: new address 23
android_work: sent uevent USB_STATE=CONNECTED
configfs-gadget gadget: high-speed config #1: b
android_work: sent uevent USB_STATE=CONFIGURED
```

The Audio Source Driver does not provide configurable kernel interfaces.

Methods to view Audio Source information:

In the output shown below, the Audio Source corresponds to card2 (audiosource) and has only one playback device node: pcmC2D0p.

```
# cat /proc/asound/cards
 0 [rockchiprk3308v]: rockchip_rk3308 - rockchip,rk3308-vad
                      rockchip,rk3308-vad
 1 [rockchiprk3308p]: rockchip_rk3308 - rockchip,rk3308-pcm
                      rockchip,rk3308-pcm
 2 [audiosource    ]: audio_source - audio_source
                      USB accessory audio source
 7 [Loopback       ]: Loopback - Loopback
                      Loopback 1

# ls -lh /proc/asound/card2
-r--r--r--    1 root     root           0 Dec 31 19:06 id
dr-xr-xr-x    3 root     root           0 Dec 31 19:06 pcm0p

# ls /dev/snd/
controlC0  controlC2  pcmC0D0c   pcmC1D0c   pcmC2D0p   pcmC7D0p   pcmC7D1p
controlC1  controlC7  pcmC0D0p   pcmC1D0p   pcmC7D0c   pcmC7D1c   timer
```

4.2 Audio Source Test

Test the Audio Source recording function:

For recording settings on a Windows PC, refer to UAC Test on Windows.

For recording settings on an Ubuntu PC, refer to UAC Test on Ubuntu.

Test commands on the RK3308 side (assuming the default sampling rate of 44.1 kHz):

```
aplay /tmp/test.wav -r 44100 -c 2 -D hw:2,0
```

Note:

- The sample rate of the test audio file `test.wav` must match the recording sample rate; otherwise, noise or silence may occur during testing.

Additionally, since the Audio Source supports 15 different sample rates, the sample rate can be dynamically configured on the PC as follows:

Open "Sound Settings" --> "Sound Control Panel" --> "Recording" --> "Properties" --> "Advanced", and select the corresponding sample rate.

See Figure 4-3 below.

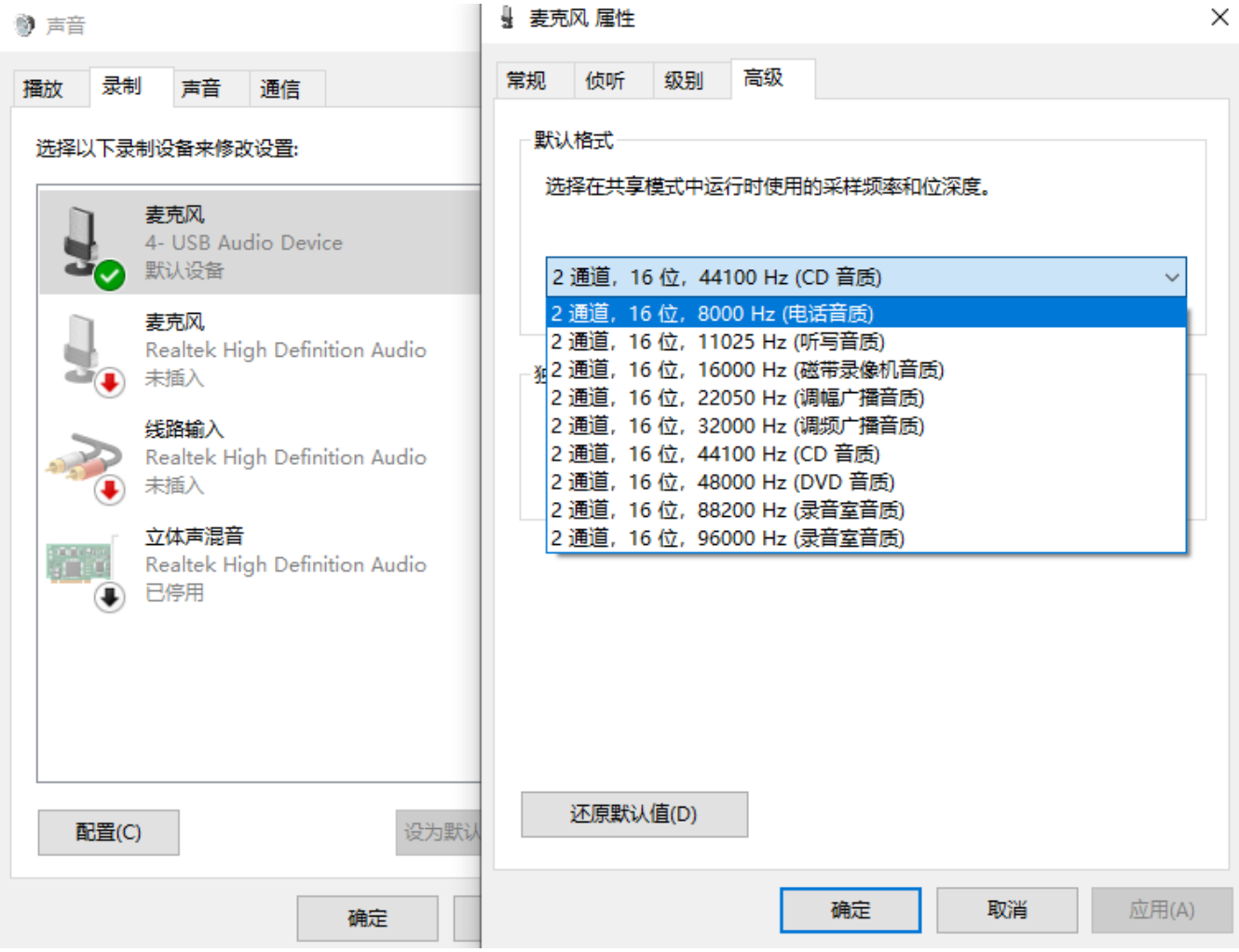


Figure 4-3 Windows-USB-Audio-Source-Setting

5. UAC1 Legacy and Audio Source Composite Usage and Test

5.1 UAC1 Legacy and Audio Source Composite Usage

Usage of UAC1 Legacy + Audio Source is as follows:

UAC1 Legacy and Audio Source can be combined into a USB composite device, supporting recording and playback functions.

Add CONFIG_USB_CONFIGFS_F_UAC1_LEGACY=y to the kernel defconfig.

Add CONFIG_USB_CONFIGFS_F_ACC=y (Audio Source depends on it) to the kernel defconfig.

Add CONFIG_USB_CONFIGFS_F_AUDIO_SRC=y to the kernel defconfig.

Additionally, the patch "**support_uac1_legacy_and_audio_source.patch**" must be applied separately.

Taking RK3308 EVB as an example:

The script for configuring UAC1 Legacy + Audio Source is referenced below:

```
mount -t configfs none /sys/kernel/config
mkdir /sys/kernel/config/usb_gadget/rockchip  -m 0770
echo 0x2207 > /sys/kernel/config/usb_gadget/rockchip/idVendor
echo 0x0019 > /sys/kernel/config/usb_gadget/rockchip/idProduct
echo 0x0100 > /sys/kernel/config/usb_gadget/rockchip/bcdDevice
mkdir /sys/kernel/config/usb_gadget/rockchip/strings/0x409   -m 0770
echo "0123456789ABCDEF" > /sys/kernel/config/usb_gadget/rockchip/strings/0x409/serialnumber
echo "rockchip"  > /sys/kernel/config/usb_gadget/rockchip/strings/0x409/manufacturer
echo "USB Audio Device"  > /sys/kernel/config/usb_gadget/rockchip/strings/0x409/product
mkdir /sys/kernel/config/usb_gadget/rockchip/configs/b.1  -m 0770
mkdir /sys/kernel/config/usb_gadget/rockchip/configs/b.1/strings/0x409  -m 0770
echo 500 > /sys/kernel/config/usb_gadget/rockchip/configs/b.1/MaxPower
echo "uac1" > /sys/kernel/config/usb_gadget/rockchip/configs/b.1/strings/0x409/configuration
mkdir /sys/kernel/config/usb_gadget/rockchip/functions/uac1_legacy.gs0
ln -s /sys/kernel/config/usb_gadget/rockchip/functions/uac1_legacy.gs0
/sys/kernel/config/usb_gadget/rockchip/configs/b.1/uac1_legacy.gs0
mkdir /sys/kernel/config/usb_gadget/rockchip/functions/audio_source.gs0
ln -s /sys/kernel/config/usb_gadget/rockchip/functions/audio_source.gs0
/sys/kernel/config/usb_gadget/rockchip/configs/b.1/audio_source.gs0
echo ff400000.usb > /sys/kernel/config/usb_gadget/rockchip/UDC
```

For other configuration and debugging methods, refer to UAC1 Legacy Usage and Audio Source Usage.

5.2 UAC1 Legacy and Audio Source Composite Test

Refer to UAC1 Legacy Test and Audio Source Test.

6. Reference Documentation

USB Protocol (from USB Implementers Forum)

- Universal Serial Bus Specification, Revision 2.0
- Universal Serial Bus Audio Device Class Specification for Basic Audio Devices
- Universal Serial Bus Device Class Definition for Audio Devices, Release 1.0
- Universal Serial Bus Device Class Definition for Audio Devices, Specification for Audio Devices, Release 2.0
- Universal Serial Bus Device Class Definition for Audio Data Formats (referred to in this document as USB Audio Data Formats)
- Universal Serial Bus Device Class Definition for Terminal Types (referred to in this document as USB Audio Terminal Types)

Others

- The Well-Tempered Computer (An introduction to computer audio) - USB
- Windows USB Audio 2.0 Drivers
- Windows Audio-Tapered Volume Controls

7. Appendix A UAC1 Device Descriptor

```
Device Descriptor:
  bLength                18
  bDescriptorType         1
  bcdUSB               2.00
  bDeviceClass            0 (Defined at Interface level)
  bDeviceSubClass         0
  bDeviceProtocol         0
  bMaxPacketSize0        64
  idVendor           0x2207
  idProduct          0x0019
  bcdDevice            1.00
  iManufacturer           1 rockchip
  iProduct                2 USB Audio Device
  iSerial                 3 0123456789ABCDEF
  bNumConfigurations      1
  Configuration Descriptor:
    bLength                 9
    bDescriptorType         2
    wTotalLength          174
    bNumInterfaces          3
    bConfigurationValue     1
    iConfiguration          4 audio
    bmAttributes         0x80
      (Bus Powered)
    MaxPower              500mA
    Interface Descriptor:
      bLength                 9
      bDescriptorType         4
      bInterfaceNumber        0
      bAlternateSetting       0
      bNumEndpoints           0
      bInterfaceClass         1 Audio
      bInterfaceSubClass      1 Control Device
      bInterfaceProtocol      0
      iInterface              5 AC Interface
      AudioControl Interface Descriptor:
        bLength                10
        bDescriptorType        36
        bDescriptorSubtype      1 (HEADER)
        bcdADC               1.00
        wTotalLength           52
        bInCollection           2
        baInterfaceNr( 0)       1
```

```
      baInterfaceNr( 1)       2
    AudioControl Interface Descriptor:
      bLength                12
      bDescriptorType        36
      bDescriptorSubtype      2 (INPUT_TERMINAL)
      bTerminalID             1
      wTerminalType      0x0101 USB Streaming
      bAssocTerminal          0
      bNrChannels             2
      wChannelConfig     0x0003
        Left Front (L)
        Right Front (R)
      iChannelNames           7 Playback Channels
      iTerminal               6 Playback Input terminal
    AudioControl Interface Descriptor:
      bLength                 9
      bDescriptorType        36
      bDescriptorSubtype      3 (OUTPUT_TERMINAL)
      bTerminalID             2
      wTerminalType      0x0301 Speaker
      bAssocTerminal          0
      bSourceID               1
      iTerminal               8 Playback Output terminal
    AudioControl Interface Descriptor:
      bLength                12
      bDescriptorType        36
      bDescriptorSubtype      2 (INPUT_TERMINAL)
      bTerminalID             3
      wTerminalType      0x0201 Microphone
      bAssocTerminal          0
      bNrChannels             2
      wChannelConfig     0x0003
        Left Front (L)
        Right Front (R)
      iChannelNames          10 Capture Channels
      iTerminal               9 Capture Input terminal
    AudioControl Interface Descriptor:
      bLength                 9
      bDescriptorType        36
      bDescriptorSubtype      3 (OUTPUT_TERMINAL)
      bTerminalID             4
      wTerminalType      0x0101 USB Streaming
      bAssocTerminal          0
      bSourceID               3
      iTerminal              11 Capture Output terminal
  Interface Descriptor:
    bLength                 9
    bDescriptorType         4
    bInterfaceNumber        1
    bAlternateSetting       0
    bNumEndpoints           0
    bInterfaceClass         1 Audio
    bInterfaceSubClass      2 Streaming
```

```
    bInterfaceProtocol      0
    iInterface             12 Playback Inactive
  Interface Descriptor:
    bLength                 9
    bDescriptorType         4
    bInterfaceNumber        1
    bAlternateSetting       1
    bNumEndpoints           1
    bInterfaceClass         1 Audio
    bInterfaceSubClass      2 Streaming
    bInterfaceProtocol      0
    iInterface             13 Playback Active
    AudioStreaming Interface Descriptor:
      bLength                 7
      bDescriptorType        36
      bDescriptorSubtype      1 (AS_GENERAL)
      bTerminalLink           1
      bDelay                  1 frames
      wFormatTag              1 PCM
    AudioStreaming Interface Descriptor:
      bLength                11
      bDescriptorType        36
      bDescriptorSubtype      2 (FORMAT_TYPE)
      bFormatType             1 (FORMAT_TYPE_I)
      bNrChannels             2
      bSubframeSize           2
      bBitResolution         16
      bSamFreqType            1 Discrete
      tSamFreq[ 0]        48000
    Endpoint Descriptor:
      bLength                 9
      bDescriptorType         5
      bEndpointAddress     0x02  EP 2 OUT
      bmAttributes            9
        Transfer Type            Isochronous
        Synch Type               Adaptive
        Usage Type               Data
      wMaxPacketSize     0x00c8  1x 200 bytes
      bInterval               4
      bRefresh                0
      bSynchAddress           0
      AudioControl Endpoint Descriptor:
        bLength                 7
        bDescriptorType        37
        bDescriptorSubtype      1 (EP_GENERAL)
        bmAttributes         0x01
          Sampling Frequency
        bLockDelayUnits         1 Milliseconds
        wLockDelay              1 Milliseconds
  Interface Descriptor:
    bLength                 9
    bDescriptorType         4
    bInterfaceNumber        2
```

```
    bAlternateSetting       0
    bNumEndpoints           0
    bInterfaceClass         1 Audio
    bInterfaceSubClass      2 Streaming
    bInterfaceProtocol      0
    iInterface             14 Capture Inactive
  Interface Descriptor:
    bLength                 9
    bDescriptorType         4
    bInterfaceNumber        2
    bAlternateSetting       1
    bNumEndpoints           1
    bInterfaceClass         1 Audio
    bInterfaceSubClass      2 Streaming
    bInterfaceProtocol      0
    iInterface             15 Capture Active
    AudioStreaming Interface Descriptor:
      bLength                 7
      bDescriptorType        36
      bDescriptorSubtype      1 (AS_GENERAL)
      bTerminalLink           4
      bDelay                  1 frames
      wFormatTag              1 PCM
    AudioStreaming Interface Descriptor:
      bLength                11
      bDescriptorType        36
      bDescriptorSubtype      2 (FORMAT_TYPE)
      bFormatType             1 (FORMAT_TYPE_I)
      bNrChannels             2
      bSubframeSize           2
      bBitResolution         16
      bSamFreqType            1 Discrete
      tSamFreq[ 0]        48000
    Endpoint Descriptor:
      bLength                 9
      bDescriptorType         5
      bEndpointAddress     0x81  EP 1 IN
      bmAttributes            5
        Transfer Type            Isochronous
        Synch Type               Asynchronous
        Usage Type               Data
      wMaxPacketSize     0x00c8  1x 200 bytes
      bInterval               4
      bRefresh                0
      bSynchAddress           0
      AudioControl Endpoint Descriptor:
        bLength                 7
        bDescriptorType        37
        bDescriptorSubtype      1 (EP_GENERAL)
        bmAttributes         0x01
          Sampling Frequency
        bLockDelayUnits         0 Undefined
        wLockDelay              0 Undefined
```

```
Device Qualifier (for other device speed):
  bLength                10
  bDescriptorType         6
  bcdUSB               2.00
  bDeviceClass            0 (Defined at Interface level)
  bDeviceSubClass         0
  bDeviceProtocol         0
  bMaxPacketSize0        64
  bNumConfigurations      1
Device Status:     0x0000
  (Bus Powered)
```

8. Appendix B UAC2 Device Descriptor

```
Device Descriptor:
  bLength                18
  bDescriptorType         1
  bcdUSB               2.00
  bDeviceClass            0 (Defined at Interface level)
  bDeviceSubClass         0
  bDeviceProtocol         0
  bMaxPacketSize0        64
  idVendor           0x2207
  idProduct          0x0019
  bcdDevice            2.00
  iManufacturer           1 rockchip
  iProduct                2 USB Audio Device
  iSerial                 3 0123456789ABCDEF
  bNumConfigurations      1
  Configuration Descriptor:
    bLength                 9
    bDescriptorType         2
    wTotalLength          219
    bNumInterfaces          3
    bConfigurationValue     1
    iConfiguration          4 audio
    bmAttributes         0x80
      (Bus Powered)
    MaxPower              500mA
    Interface Association:
      bLength                 8
      bDescriptorType        11
      bFirstInterface         0
      bInterfaceCount         3
      bFunctionClass          1 Audio
      bFunctionSubClass       0
      bFunctionProtocol      32
      iFunction               5 Source/Sink
    Interface Descriptor:
```

```
    bLength                 9
    bDescriptorType         4
    bInterfaceNumber        0
    bAlternateSetting       0
    bNumEndpoints           0
    bInterfaceClass         1 Audio
    bInterfaceSubClass      1 Control Device
    bInterfaceProtocol     32
    iInterface              6 Topology Control
    AudioControl Interface Descriptor:
      bLength                 9
      bDescriptorType        36
      bDescriptorSubtype      1 (HEADER)
      bcdADC               2.00
      bCategory               8
      wTotalLength           83
      bmControl            0x00
    AudioControl Interface Descriptor:
      bLength                 8
      bDescriptorType        36
      bDescriptorSubtype     10 (CLOCK_SOURCE)
      bClockID                6
      bmAttributes         0x01 Internal fixed Clock
      bmControls           0x01
        Clock Frequency Control (read-only)
      bAssocTerminal          0
      iClockSource            7 48000Hz
    AudioControl Interface Descriptor:
      bLength                 8
      bDescriptorType        36
      bDescriptorSubtype     10 (CLOCK_SOURCE)
      bClockID                5
      bmAttributes         0x01 Internal fixed Clock
      bmControls           0x01
        Clock Frequency Control (read-only)
      bAssocTerminal          0
      iClockSource            8 64000Hz
    AudioControl Interface Descriptor:
      bLength                17
      bDescriptorType        36
      bDescriptorSubtype      2 (INPUT_TERMINAL)
      bTerminalID             1
      wTerminalType      0x0101 USB Streaming
      bAssocTerminal          0
      bCSourceID              5
      bNrChannels             2
      bmChannelConfig   0x00000003
        Front Left (FL)
        Front Right (FR)
      bmControls    0x0003
        Copy Protect Control (read/write)
      iChannelNames           0
      iTerminal               9 USBH Out
```

```
    AudioControl Interface Descriptor:
      bLength                17
      bDescriptorType        36
      bDescriptorSubtype      2 (INPUT_TERMINAL)
      bTerminalID             2
      wTerminalType      0x0201 Microphone
      bAssocTerminal          0
      bCSourceID              6
      bNrChannels             2
      bmChannelConfig   0x00000003
        Front Left (FL)
        Front Right (FR)
      bmControls    0x0003
        Copy Protect Control (read/write)
      iChannelNames           0
      iTerminal              10 USBD Out
    AudioControl Interface Descriptor:
      bLength                12
      bDescriptorType        36
      bDescriptorSubtype      3 (OUTPUT_TERMINAL)
      bTerminalID             4
      wTerminalType      0x0101 USB Streaming
      bAssocTerminal          0
      bSourceID               2
      bCSourceID              6
      bmControls         0x0003
        Copy Protect Control (read/write)
      iTerminal              11 USBH In
    AudioControl Interface Descriptor:
      bLength                12
      bDescriptorType        36
      bDescriptorSubtype      3 (OUTPUT_TERMINAL)
      bTerminalID             3
      wTerminalType      0x0301 Speaker
      bAssocTerminal          0
      bSourceID               1
      bCSourceID              5
      bmControls         0x0003
        Copy Protect Control (read/write)
      iTerminal              12 USBD In
  Interface Descriptor:
    bLength                 9
    bDescriptorType         4
    bInterfaceNumber        1
    bAlternateSetting       0
    bNumEndpoints           0
    bInterfaceClass         1 Audio
    bInterfaceSubClass      2 Streaming
    bInterfaceProtocol     32
    iInterface             13 Playback Inactive
  Interface Descriptor:
    bLength                 9
    bDescriptorType         4
```

```
    bInterfaceNumber        1
    bAlternateSetting       1
    bNumEndpoints           1
    bInterfaceClass         1 Audio
    bInterfaceSubClass      2 Streaming
    bInterfaceProtocol     32
    iInterface             14 Playback Active
    AudioStreaming Interface Descriptor:
      bLength                16
      bDescriptorType        36
      bDescriptorSubtype      1 (AS_GENERAL)
      bTerminalLink           1
      bmControls           0x00
      bFormatType             1
      bmFormats         0x00000001
        PCM
      bNrChannels             2
      bmChannelConfig   0x00000003
        Front Left (FL)
        Front Right (FR)
      iChannelNames           0
    AudioStreaming Interface Descriptor:
      bLength                 6
      bDescriptorType        36
      bDescriptorSubtype      2 (FORMAT_TYPE)
      bFormatType             1 (FORMAT_TYPE_I)
      bSubslotSize            2
      bBitResolution         16
    Endpoint Descriptor:
      bLength                 7
      bDescriptorType         5
      bEndpointAddress     0x02  EP 2 OUT
      bmAttributes            9
        Transfer Type            Isochronous
        Synch Type               Adaptive
        Usage Type               Data
      wMaxPacketSize     0x0100  1x 256 bytes
      bInterval               4
      AudioControl Endpoint Descriptor:
        bLength                 8
        bDescriptorType        37
        bDescriptorSubtype      1 (EP_GENERAL)
        bmAttributes         0x00
        bmControls           0x00
        bLockDelayUnits         0 Undefined
        wLockDelay              0
  Interface Descriptor:
    bLength                 9
    bDescriptorType         4
    bInterfaceNumber        2
    bAlternateSetting       0
    bNumEndpoints           0
    bInterfaceClass         1 Audio
```

```
      bInterfaceSubClass      2 Streaming
      bInterfaceProtocol     32
      iInterface             15 Capture Inactive
    Interface Descriptor:
      bLength                 9
      bDescriptorType         4
      bInterfaceNumber        2
      bAlternateSetting       1
      bNumEndpoints           1
      bInterfaceClass         1 Audio
      bInterfaceSubClass      2 Streaming
      bInterfaceProtocol     32
      iInterface             16 Capture Active
      AudioStreaming Interface Descriptor:
        bLength                16
        bDescriptorType        36
        bDescriptorSubtype      1 (AS_GENERAL)
        bTerminalLink           4
        bmControls           0x00
        bFormatType             1
        bmFormats         0x00000001
          PCM
        bNrChannels             2
        bmChannelConfig   0x00000003
          Front Left (FL)
          Front Right (FR)
        iChannelNames           0
      AudioStreaming Interface Descriptor:
        bLength                 6
        bDescriptorType        36
        bDescriptorSubtype      2 (FORMAT_TYPE)
        bFormatType             1 (FORMAT_TYPE_I)
        bSubslotSize            2
        bBitResolution         16
      Endpoint Descriptor:
        bLength                 7
        bDescriptorType         5
        bEndpointAddress     0x81  EP 1 IN
        bmAttributes           13
          Transfer Type            Isochronous
          Synch Type               Synchronous
          Usage Type               Data
        wMaxPacketSize     0x00c0  1x 192 bytes
        bInterval               4
        AudioControl Endpoint Descriptor:
          bLength                 8
          bDescriptorType        37
          bDescriptorSubtype      1 (EP_GENERAL)
          bmAttributes         0x00
          bmControls           0x00
          bLockDelayUnits         0 Undefined
          wLockDelay              0
Device Qualifier (for other device speed):
```

```
  bLength                10
  bDescriptorType         6
  bcdUSB               2.00
  bDeviceClass            0 (Defined at Interface level)
  bDeviceSubClass         0
  bDeviceProtocol         0
  bMaxPacketSize0        64
  bNumConfigurations      1
Device Status:     0x0000
  (Bus Powered)
```